O'REILLY

Deep Learning for Business Made Simple

Dr. Chester Ismay

Elevator Pitch

- A hands-on, business-first introduction to deep learning -- no heavy math required
- Build and interpret neural networks for images, text, and structured business data with Keras 3 (PyTorch backend)
- Use Generative AI to write code, debug models, and explain results
- Leave able to spot where deep learning creates real business value -- and communicate it

Meet your Instructor

Dr. Chester Ismay (please call me “Chet”)

- Data science educator and AI consultant
- PhD in Statistics
- Co-author of “Statistical Inference via Data Science: A ModernDive into R and the Tidyverse” (moderndive.com/v2/)
- Background across academia, online education, corporate training, bootcamps, and consulting
- Been to all 50 US States

Learning Objectives

- By the end of this course, you will be able to:
    - Understand deep learning's building blocks and their business uses
    - Build, train, and evaluate neural networks with Keras 3 (PyTorch backend)
    - Prepare business data with preprocessing and feature extraction
    - Use Generative AI to assist with coding, debugging, and interpretation
    - Explain and visualize results for non-technical audiences
    - Recognize when deep learning offers a strategic advantage

Course schedule

- Intro: Deep Learning in the Modern Business Landscape
- Module 1: Demystifying Neural Networks for Business Insight
- Module 2: Deep Learning for Images and Customer Experience
- Module 3: Deep Learning for Text and Customer Sentiment
- Module 4: Interpreting and Communicating Deep Learning Results
- Wrap-Up and Review

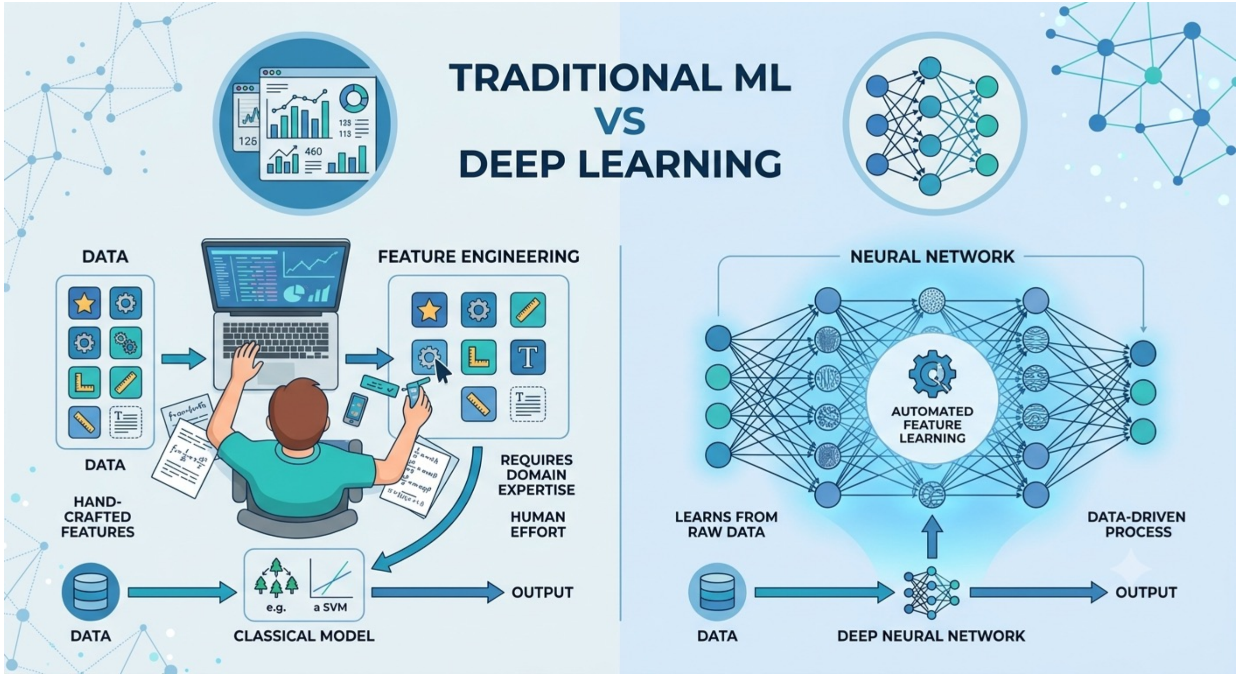
Poll in ON24 (to be entered by O’Reilly team)

Which deep learning application are you most eager to explore today?

- A. Image classification and visual insights
- B. Text and customer sentiment
- C. Predicting outcomes from business data
- D. Interpreting and communicating model results

Intro:
Deep Learning in the Modern Business Landscape

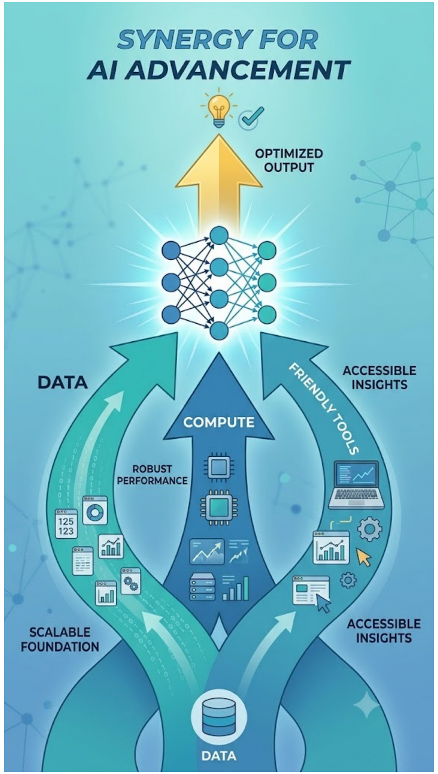
What Is Deep Learning?



- The AI chat assistants you already use -- large language models (LLMs) -- are deep learning, too!

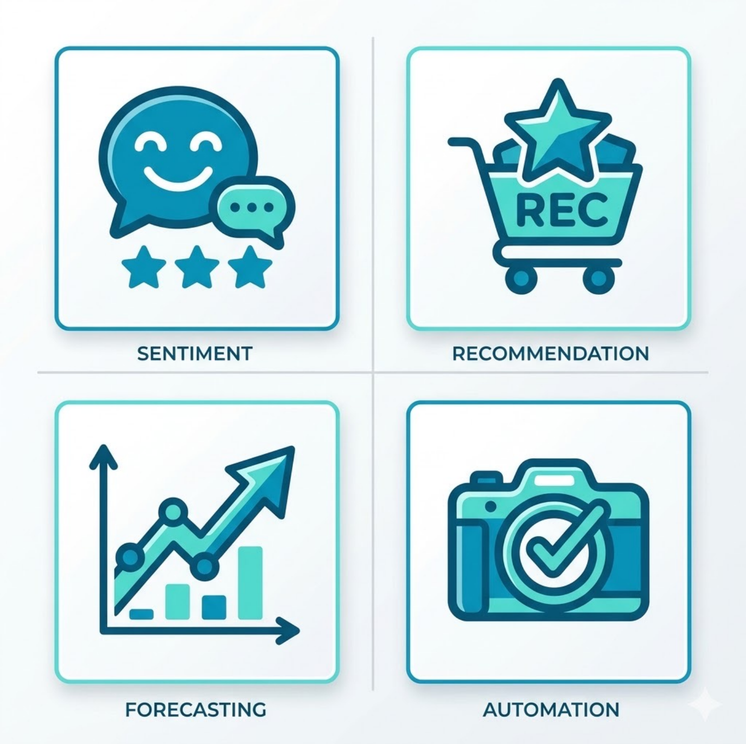
Why Deep Learning Now?

- More data than ever from apps, sensors, and customers
- Accessible compute -- a laptop or a free cloud notebook is enough to start
- High-level frameworks (Keras 3, PyTorch) remove the math-heavy barrier



Where Deep Learning Creates Business Value

- Customer insight -- sentiment from reviews and support logs
- Product recommendation and personalization
- Trend and demand forecasting
- Automation -- image tagging, document processing, defect detection



Walkthrough + Exercise 0.1

Setting Up the Python Environment with Keras 3, PyTorch, and Colab

- By the end of this exercise, you will be able to:
  - Open and configure a Google Colab / Jupyter notebook for deep learning
  - Install and import Keras 3 (PyTorch backend)
  - Verify the environment with a quick tensor operation
  - Use a GenAI assistant to resolve setup or import errors
- Tip: follow along in the live notebook -- we will screen-share it.

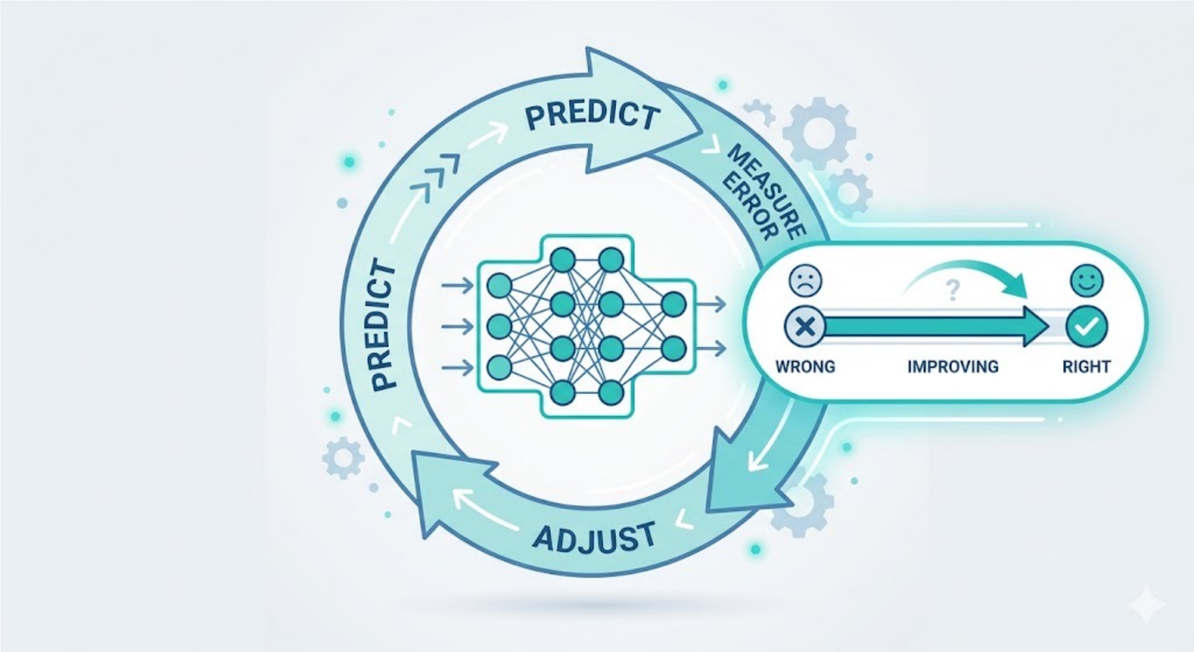
Q&A

Intro

Module 1:
Demystifying Neural Networks for Business Insight

What Is a Neural Network?

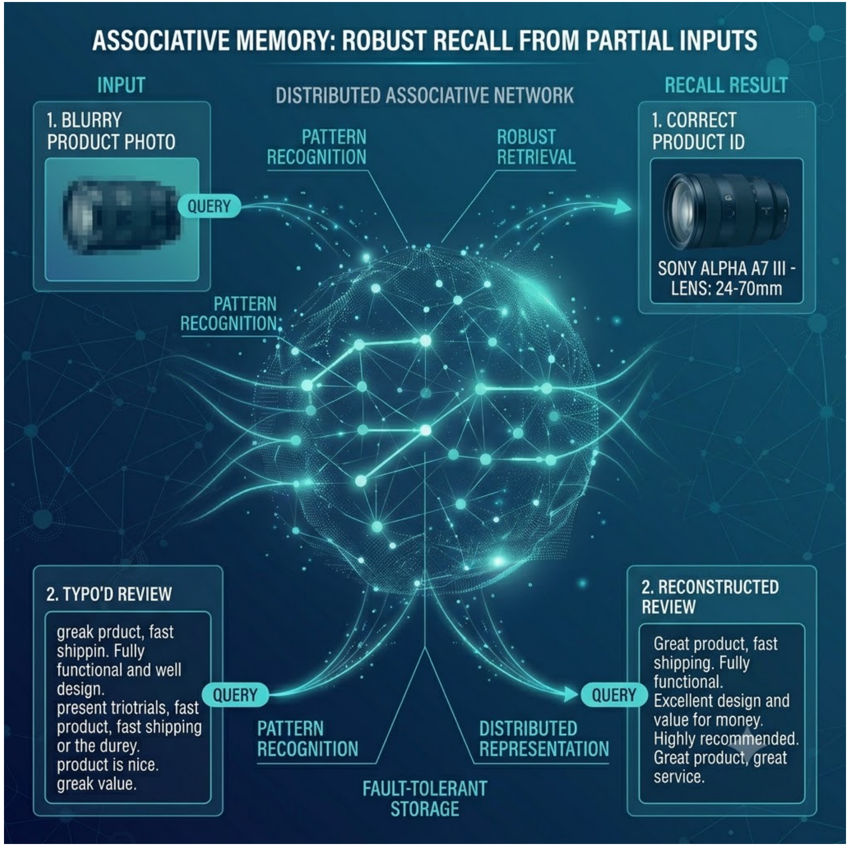
How Neural Networks Learn



- Scale these same ideas to billions of connections trained on massive text, and you get an LLM!

How Neural Networks “Remember”

- Knowledge is spread across the whole network, not stored at one address
- Recall works from partial or fuzzy cues -- like remembering a face from a name
- Robust to noise and small errors, so messy real-world inputs still work
- Unlike a database, there is no exact key to look up



Walkthrough + Exercise 1.1

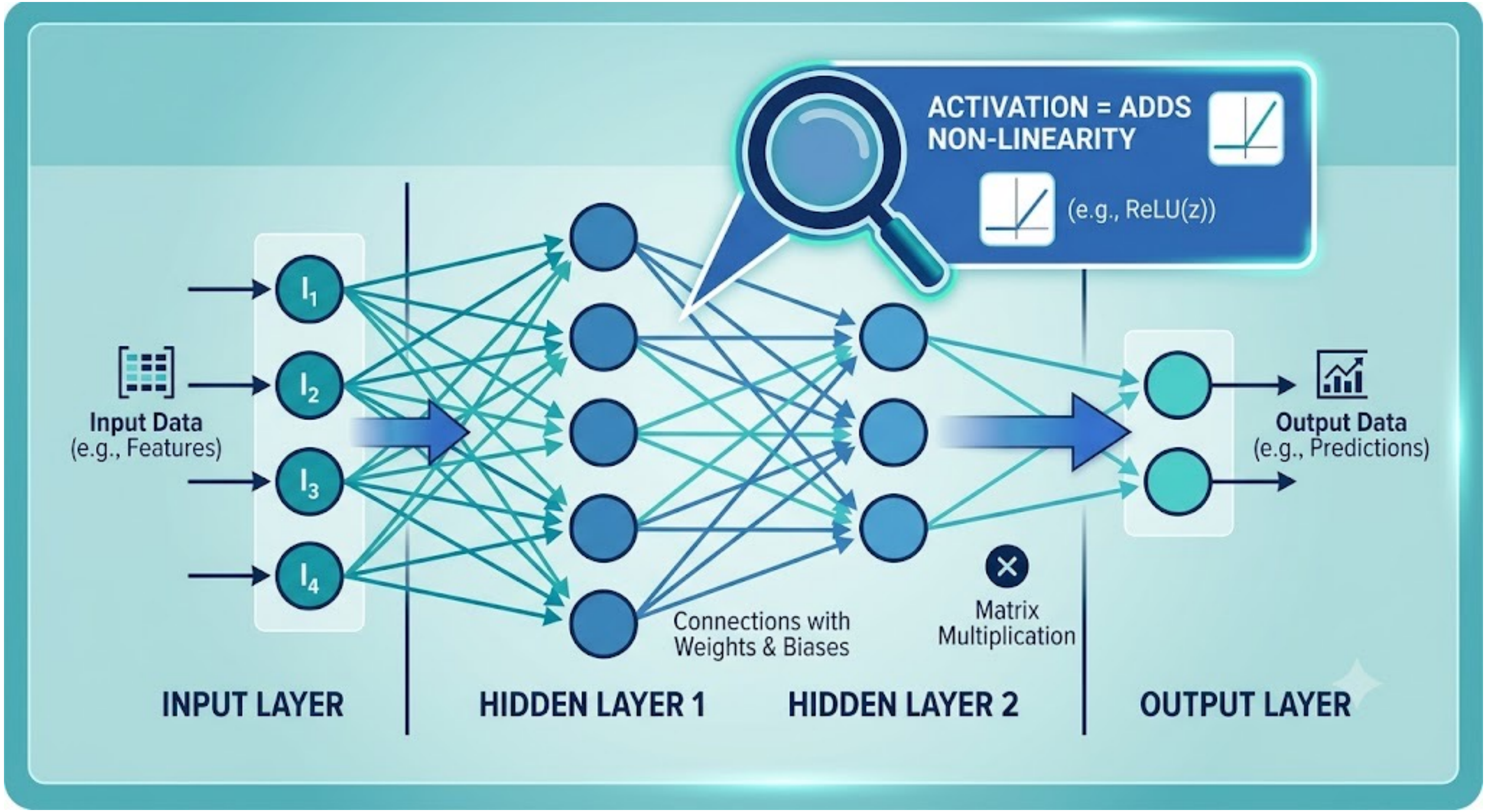
Building a Simple Feedforward Neural Network in Keras (PyTorch backend)

- By the end of this exercise, you will be able to:
    - Build a feedforward (dense) network with the Keras Sequential API
    - Predict a business outcome such as a booking cancellation or price
    - Compile the model with a loss function and optimizer, then fit it
    - Read the training output to gauge whether the model is learning
- Tip: follow along in the live notebook -- we will screen-share it.

Data Dictionary: The Hotel Booking Columns

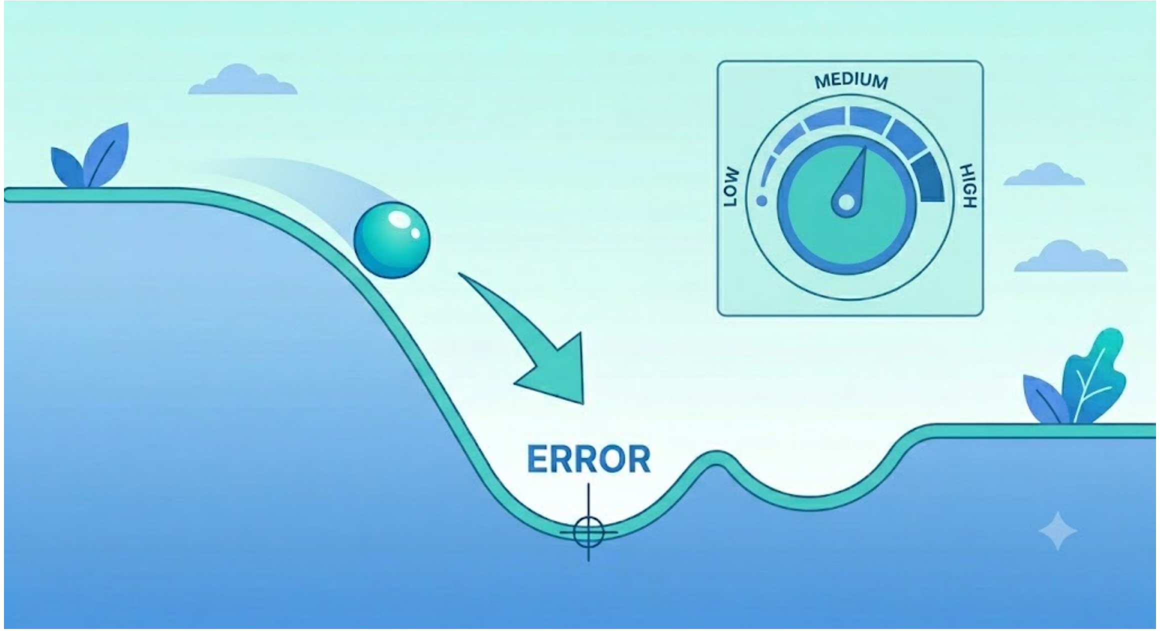
Column	Type	What it stores
hotel	text	Property type: City Hotel or Resort Hotel
lead_time	integer	Days between the booking and the arrival date
stays_in_weekend_nights	integer	Weekend nights (Sat/Sun) in the stay
stays_in_week_nights	integer	Weeknights (Mon to Fri) in the stay
adults	integer	Number of adults on the booking
previous_cancellations	integer	Earlier bookings this guest had canceled
booking_changes	integer	Changes made to the booking after it was placed
total_of_special_requests	integer	Count of special requests (e.g., a high floor)
adr	float	Average daily rate: the average price per night
is_canceled	binary (0/1)	Target: 1 if the booking was canceled, 0 if kept

Model Components, Part 1: Layers and Activations



Model Components, Part 2: Loss and Optimization

- The loss function is the business-aligned definition of “wrong”
- The optimizer and learning rate control how fast and stably it improves
- The real goal: low error on NEW data, not just the training set



Walkthrough + Exercise 1.2

Using Generative AI to Interpret Keras/PyTorch Errors and Clarify Model Behavior

- By the end of this exercise, you will be able to:
  - Paste a Keras or PyTorch error with context and get a plain-English explanation
  - Ask a GenAI assistant to explain a layer or a hyperparameter choice
  - Iterate prompts to debug shape mismatches and dtype issues
  - Validate the suggested fix by re-running the cell
- Tip: follow along in the live notebook -- we will screen-share it.

Q&A

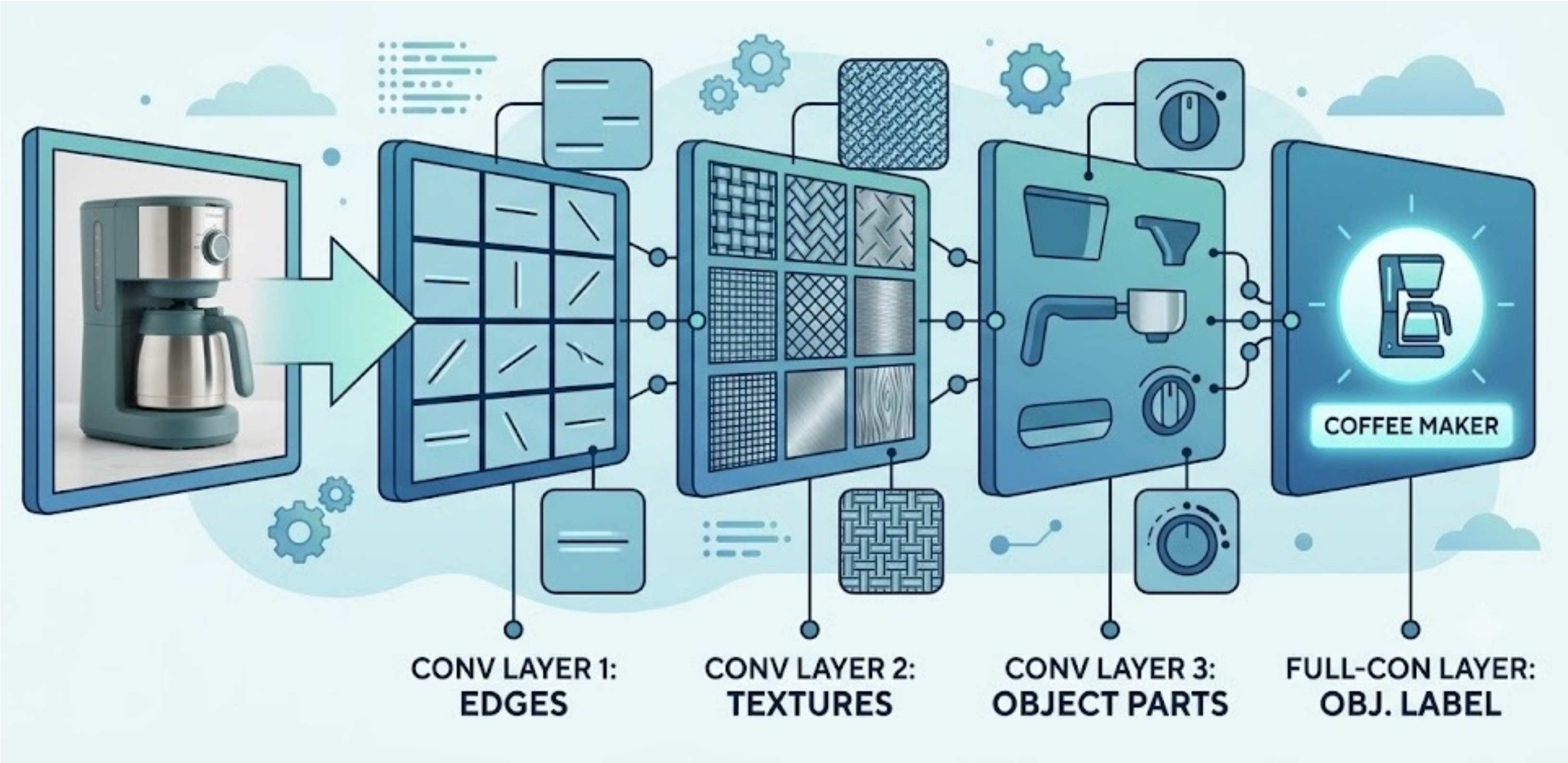
Module 1

Break

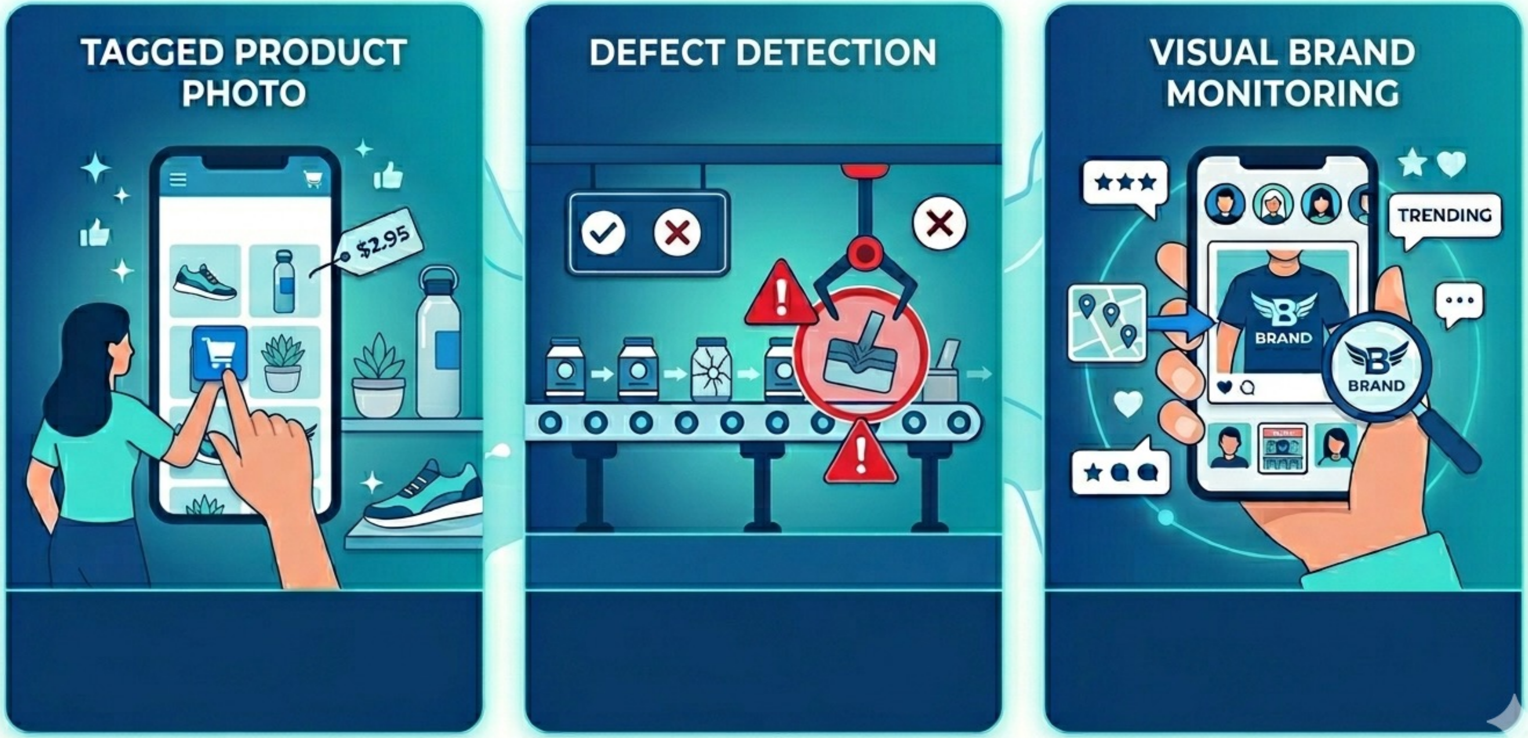
5 minutes

Module 2:
Deep Learning for Images and Customer Experience

How CNNs See: From Pixels to Patterns



CNNs in Business



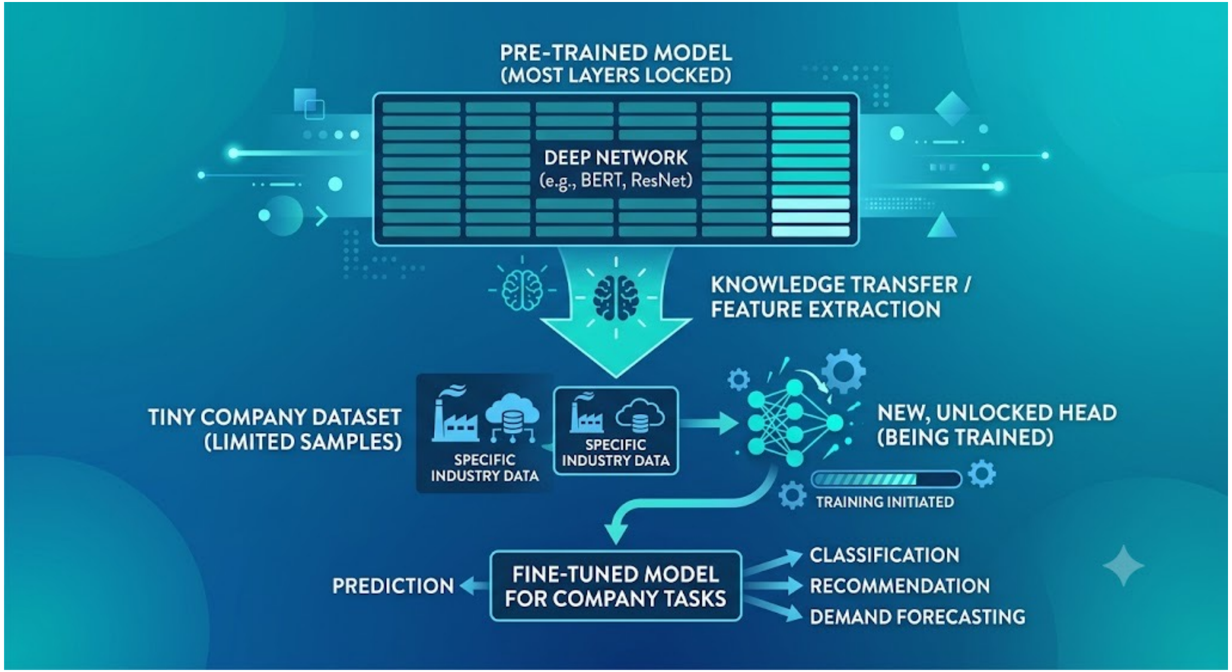
Walkthrough + Exercise 2.1

Classifying Product Images Using a Pre-Trained CNN in Keras

- By the end of this exercise, you will be able to:
    - Load a pre-trained CNN (for example MobileNet or ResNet) from Keras
    - Preprocess and classify sample product images
    - Interpret the top predicted labels and their confidence
    - Discuss business uses of automated image tagging
- Tip: follow along in the live notebook -- we will screen-share it.

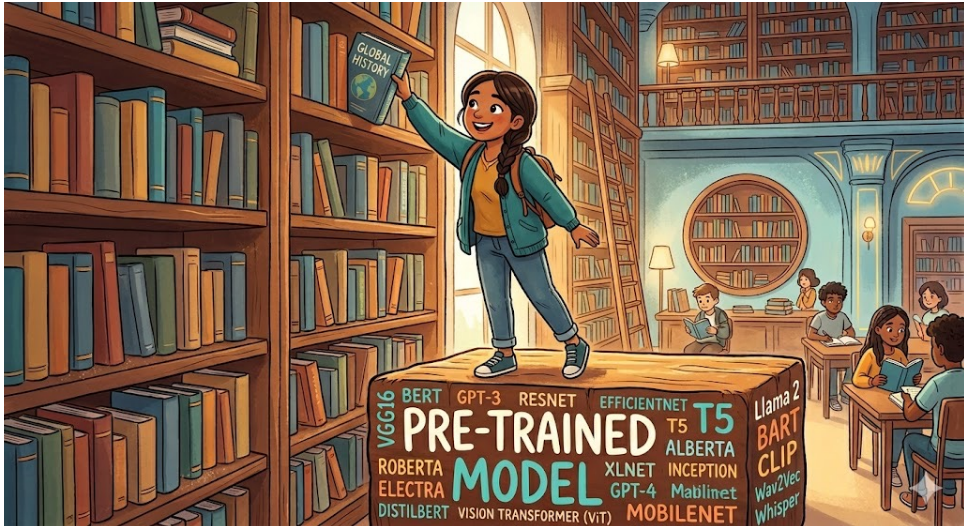
Transfer Learning: Reuse, Don’t Reinvent

- Start from a model already trained on millions of images
- Keep its general visual knowledge; retrain only a small new part
- Fine-tune on your own, much smaller, business dataset



Why Transfer Learning Matters for Business

- Far less data and compute needed -- faster time to value
- Makes deep learning feasible for smaller organizations
- Lowers the cost and risk of trying an AI project
- LLMs are the ultimate example -- a foundation model pre-trained once, then adapted to many tasks



Walkthrough + Exercise 2.2

Using Generative AI to Explain Model Layers and Suggest Improvements

- By the end of this exercise, you will be able to:
    - Ask a GenAI assistant to summarize what each layer of a model does
    - Request architecture suggestions for your specific use case
    - Compare trade-offs: accuracy vs. model size vs. speed
    - Apply and test one suggested change
- Tip: follow along in the live notebook -- we will screen-share it.

Q&A

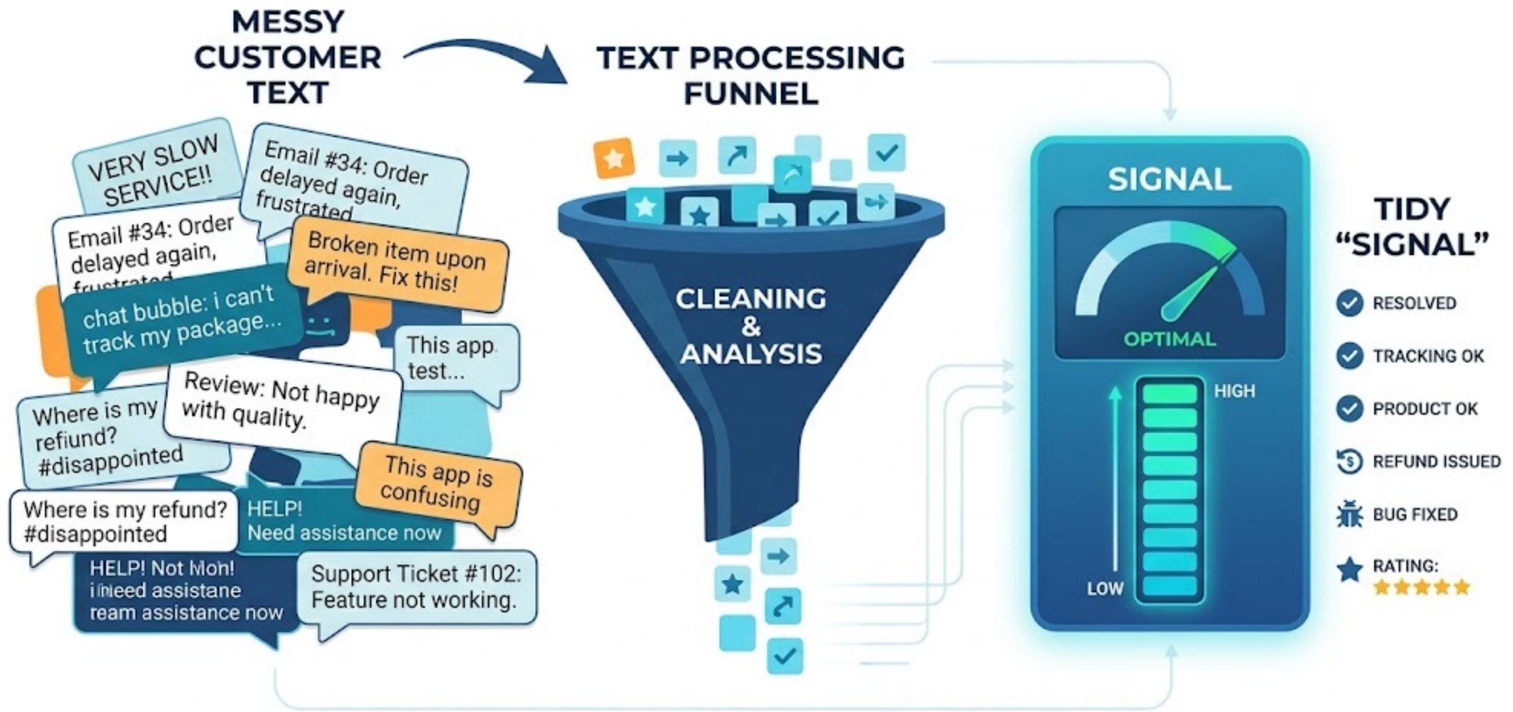
Module 2

Break

5 minutes

Module 3:
Deep Learning for Text and Customer Sentiment

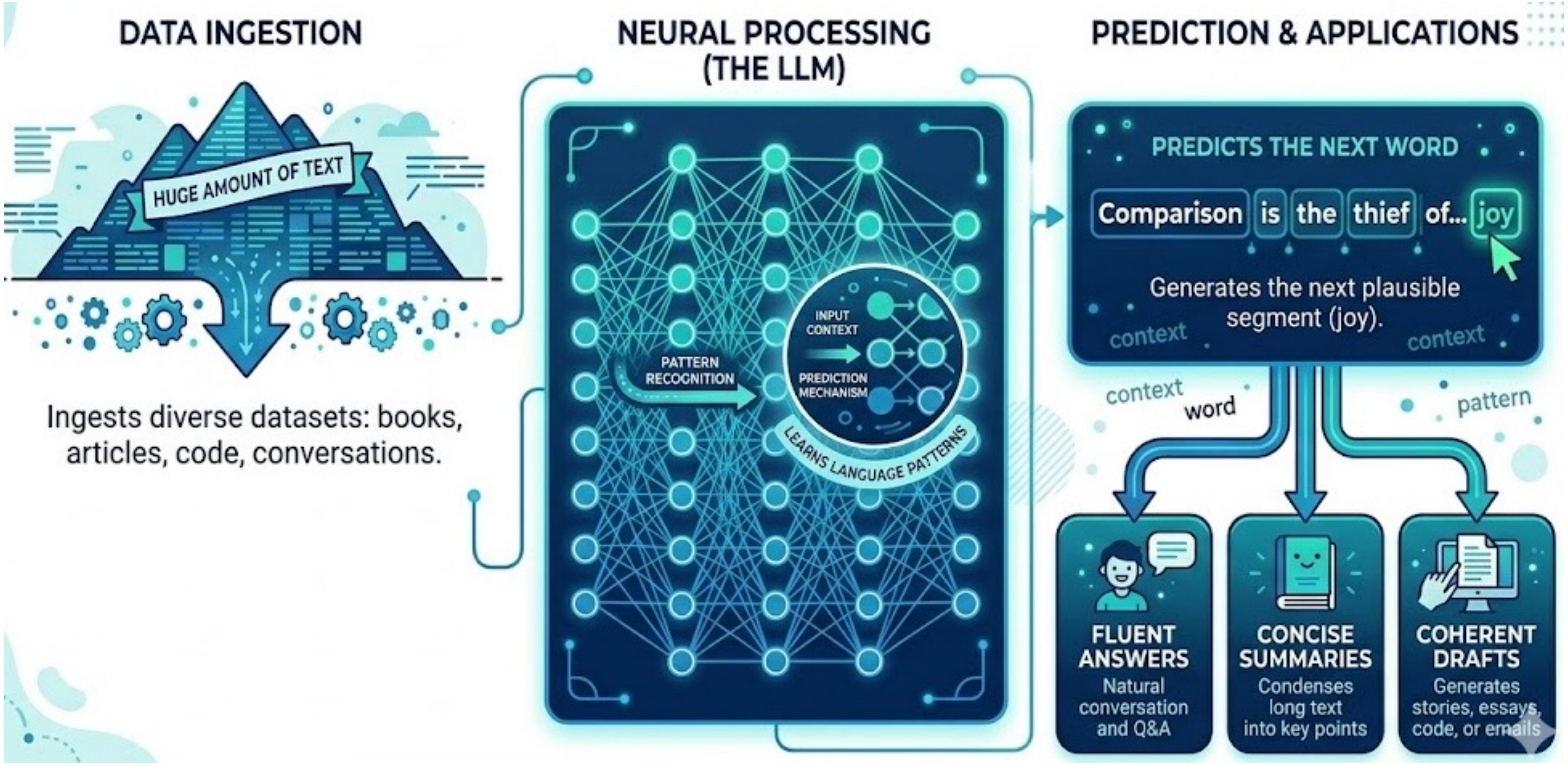
Teaching Machines to Read



Word Embeddings: Meaning as Numbers

- Words become points in space, so a model can do math on meaning
- Words with similar meaning sit close together
- Captures context -- “not great” reads very differently from “great”

Large Language Models (LLMs): NLP at Scale



NLP in Business

- Customer sentiment from reviews and surveys
- Triage and themes from support logs and tickets
- Brand and product feedback monitored at scale

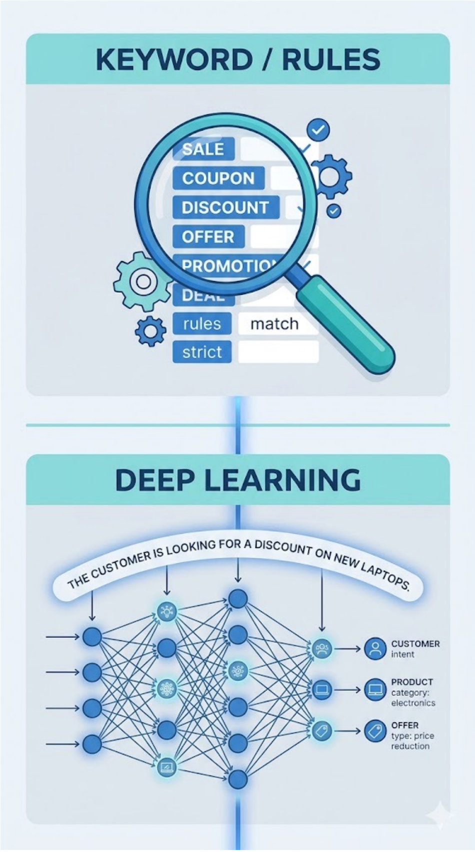
Walkthrough + Exercise 3.1

Building a Sentiment Analysis Model with Embeddings in Keras

- By the end of this exercise, you will be able to:
    - Prepare and tokenize text data
    - Add an embedding layer to a Keras model
    - Train the model to classify sentiment (positive vs. negative)
    - Evaluate performance on held-out reviews
- Tip: follow along in the live notebook -- we will screen-share it.

Traditional Analytics vs. Deep Learning for Text

- Rule and keyword methods: fast and transparent, but brittle
- Deep learning: captures context, nuance, and even sarcasm
- You can also prompt a pre-trained LLM -- often no training data or model-building needed
- Choosing the right tool: cost, data volume, and interpretability



Walkthrough + Exercise 3.2

Using Generative AI to Summarize Sentiment Results for Stakeholders

- By the end of this exercise, you will be able to:
    - Feed model outputs and metrics to a GenAI assistant
    - Generate a concise, plain-English summary
    - Tailor the tone for executives versus analysts
    - Fact-check and refine the narrative
- Tip: follow along in the live notebook -- we will screen-share it.

Q&A

Module 3

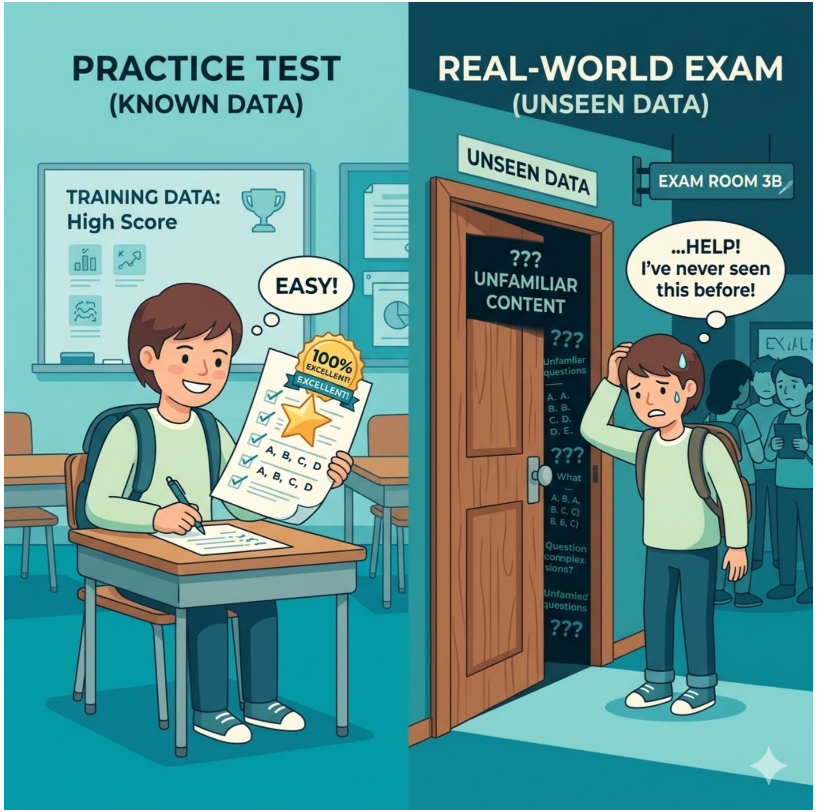
Break

5 minutes

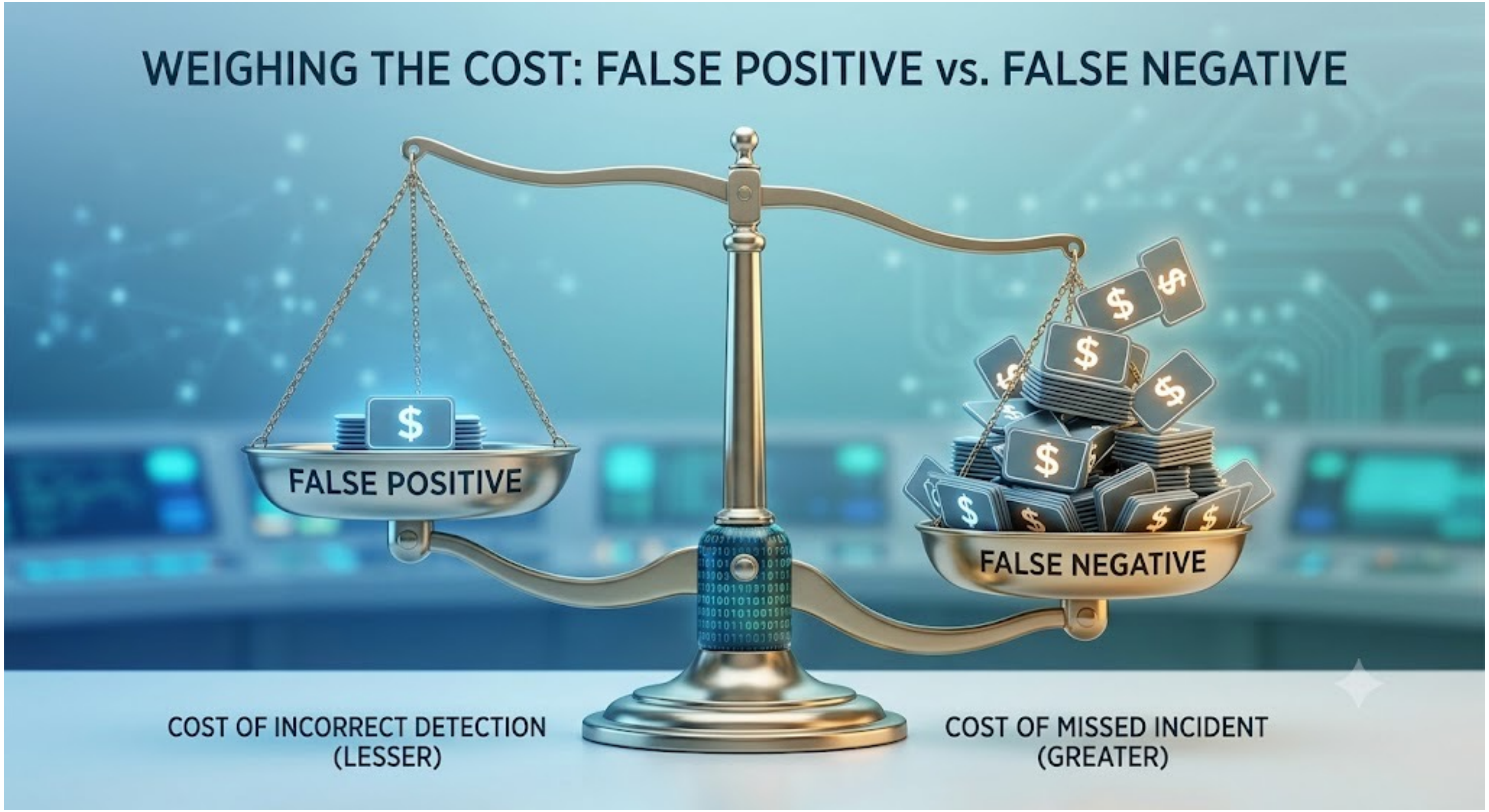
Module 4:
Interpreting and Communicating Deep Learning Results

Beyond Accuracy: Metrics That Matter

- Judge models on unseen data -- one that memorizes the training set fails in production
- Accuracy alone misleads, especially with imbalanced data
- Use precision, recall, and F1 to see what the model gets right and wrong



Aligning Metrics with Business Costs



Walkthrough + Exercise 4.1

Interpreting Predictions and Visualizing Performance

- By the end of this exercise, you will be able to:
    - Generate predictions on validation data
    - Build and read a confusion matrix
    - Visualize performance with clear, simple charts
    - Translate metrics into business implications
- Tip: follow along in the live notebook -- we will screen-share it.

Opening the Black Box

- Show “how the model thinks” with concrete examples
- Feature importance and attention maps highlight what drove a prediction
- Turn model internals into reasons a stakeholder can follow

Telling the Story for Non-Technical Audiences

- Lead with the decision and its impact, then support it with evidence
- Prefer visuals over jargon and raw numbers
- Use AI-assisted summaries to draft a clear executive brief

Walkthrough + Exercise 4.2

From Feature Importance to an AI-Drafted Executive Summary

- By the end of this exercise, you will be able to:
  - See which features drove the model (feature importance)
  - Turn the results and the why into a short executive brief
  - Highlight business impact, risks, and recommended actions
  - Review the output for accuracy and bias
- Tip: follow along in the live notebook -- we will screen-share it.

Q&A

Module 4

Wrap-Up and Review: Bringing It All Together

Deep Learning Myths vs. Reality (1 of 2)

- “It needs heavy math/CS”
  - Frameworks abstract most of the complexity
- “Neural nets are black boxes”
  - Explainability tools make them interpretable
- “Bigger models are always better”
  - Beware overfitting and diminishing returns

Deep Learning Myths vs. Reality (2 of 2)

- “Deep learning finds insights on its own”
  - Human and domain expertise are essential
- “Good prompting is just typing a question”
  - It is a skill: clarity, specificity, iteration
- “LLMs know what is truth”
  - They predict plausible text, so you need to verify facts and sources
- “You need GPUs and big data”
  - Transfer learning makes it accessible

Discussion: From Myths to Your Workflow

- Which deep learning myth surprised you most -- and why?
- Where in your organization could images, text, or structured-data models add value?
- Share one idea in the chat; we'll discuss a few together.

Takeaways

- Build, train, and read deep learning models with Keras 3 (PyTorch backend)
- Transfer learning makes deep learning practical on modest data and hardware
- Generative AI assists coding, debugging, and explanation -- always validate it
- The value is in interpretation and communication, not just accuracy
- Keep going:
  - Keras 3 and PyTorch documentation
  - Book: Deep Learning with Python, Third Edition (Watson and Chollet)
  - Book: Deep Learning for Coders with fastai and PyTorch (free and paid)
  - For the mathematically curious (free): D. MacKay, Information Theory, Inference, and Learning Algorithms -- inference.org.uk/itprnn/book.pdf
  - Prefer TensorFlow? Its docs are at tensorflow.org

Thank you!

Integrate deep learning into your organization's analytics practice.

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